

		c_j	0	0	-1	-1
c_B	B	b	a_1	a_2	a_3	a_4
-1	a_3	6	4	2	1	0
-1	a_4	4	1	5	0	1
$z_j - c_j$			-5	-7	0	0
-1	a_3	$\frac{22}{5}$	$\frac{18}{5}$	0	1	$-\frac{2}{5}$
0	a_2	$\frac{4}{5}$	$\frac{1}{5}$	1	0	$\frac{1}{5}$
$z_j - c_j$			$-\frac{18}{5}$	0	0	$\frac{7}{5}$
0	a_1	$\frac{11}{9}$	1	0	$\frac{5}{18}$	$-\frac{1}{9}$
0	a_2	$\frac{5}{9}$	0	1	$-\frac{1}{18}$	$\frac{2}{9}$
$z_j - c_j$			0	0	1	1

In the second table, $(z_1 - c_1)$ is the only negative index. Hence a_1 enters the basis and a_3 leaves it. We transform the table according to the simplex rules. In the last table, all $z_j - c_j \geq 0$. Thus the optimal solution of the L. P. P. is obtained and the vectors a_1, a_2 are present in the basis. Thus the inverse of the required matrix is directly read off from the optimum simplex tableau. This consists of the columns in the last iteration which contained the initial basis in the first table.

Thus, if

$$A = (a_1, a_2) = \begin{bmatrix} 4 & 2 \\ 1 & 5 \end{bmatrix}, \text{ then } A^{-1} = \begin{bmatrix} \frac{5}{18} & -\frac{1}{9} \\ -\frac{1}{18} & \frac{2}{9} \end{bmatrix}.$$

EXAMPLES V (B)

Use the two phase simplex method to solve the following linear programming problems (1 - 13) :

01 Maximize $z = 5x_1 + 3x_2$
subject to $3x_1 + 5x_2 \leq 15,$
 $5x_1 + 2x_2 \leq 10,$
 $x_1, x_2 \geq 0.$

[Ans. $x_1 = \frac{20}{19}, x_2 = \frac{45}{19}, z_{\max} = \frac{235}{19}.$]

02 Maximize $z = 3x_1 + 2x_2$
subject to $2x_1 + x_2 \leq 40,$
 $x_1 + x_2 \leq 24,$
 $2x_1 + 3x_2 \leq 60,$
 $x_1, x_2 \geq 0.$

[Ans. $x_1 = 16, x_2 = 8; z_{\max} = 64.$]

[Kalyani M. Sc., 1993]



113 Minimize $z = x_1 + x_2$
 subject to $2x_1 + x_2 \geq 4$,
 $x_1 + 7x_2 \geq 7$,
 $x_1, x_2 \geq 0$.

[Calcutta Hons., 1985]

[Ans. $x_1 = \frac{7}{13}$, $x_2 = \frac{13}{13}$, $z_{\min} = \frac{20}{13}$.]

114 Minimize $z = 3x_1 + 2x_2$
 subject to $2x_1 + x_2 \geq 14$,
 $2x_1 + 3x_2 \geq 22$,
 $x_1 + x_2 \geq 1$,
 $x_1, x_2 \geq 0$.

[Jadavpur, 1988]

[Ans. $x_1 = 5$, $x_2 = 4$; $z_{\min} = 23$.]

115 Minimize $z = 4x_1 + x_2$
 subject to $x_1 + 2x_2 \leq 3$,
 $4x_1 + 3x_2 \geq 6$,
 $3x_1 + x_2 = 3$,
 $x_1, x_2 \geq 0$.

[Calcutta Hons., 1991]

[Ans. $x_1 = \frac{3}{5}$, $x_2 = \frac{6}{5}$, $x_3 = 0$; $z_{\min} = \frac{18}{5}$.]

116 Maximize $z = 5x_1 + 8x_2$
 subject to $3x_1 + 2x_2 \geq 3$,
 $x_1 + 4x_2 \geq 4$,
 $x_1 + x_2 \leq 5$,
 $x_1, x_2 \geq 0$.

Use Big M-method and verify the result obtained.

[Roorkee, 1977]

[Ans. $x_1 = 0$, $x_2 = 5$; $z_{\max} = 40$.]

117 Minimize $z = 2x_1 + 9x_2 + x_3$
 subject to $x_1 + 4x_2 + 2x_3 \geq 5$,
 $3x_1 + x_2 + 2x_3 \geq 4$,
 $x_1, x_2, x_3 \geq 0$.

[Meerut, 1972]

[Ans. $x_1 = x_2 = 0$, $x_3 = \frac{5}{2}$; $z_{\min} = \frac{5}{2}$.]

118 Maximize $z = 5x_1 - 2x_2 + 3x_3$
 subject to $2x_1 + 2x_2 - x_3 \geq 2$,
 $3x_1 - 4x_2 \leq 3$,
 $x_2 + 3x_3 \leq 5$,
 $x_1, x_2, x_3 \geq 0$.

[Kalyani Hons., 1982]

[Ans. $x_1 = \frac{2}{3}$, $x_2 = 5$, $x_3 = 0$; $z_{\max} = \frac{86}{3}$.]

119 Minimize $z = x_1 - 3x_2 + 2x_3$
 subject to $3x_1 - x_2 + 2x_3 \leq 7$,
 $-2x_1 + 4x_2 \leq 12$,
 $-4x_1 + 3x_2 + 8x_3 \leq 10$,
 $x_1, x_2, x_3 \geq 0$.

[Ans. $x_1 = 4$, $x_2 = 5$, $x_3 = 0$; $z_{\min} = -11$.]

110 Maximize $z = x_1 + 2x_2 + 3x_3 - x_4$
 subject to $x_1 + 2x_2 + 3x_3 = 15$,
 $2x_1 + x_2 + 5x_3 = 20$,
 $x_1 + 2x_2 + x_3 + x_4 = 10$,
 $x_1, x_2, x_3, x_4 \geq 0$.

[Ans. $x_1 = 0$, $x_2 = \frac{15}{7}$, $x_3 = \frac{25}{7}$, $x_4 = 0$; $z_{\max} = 15$.]

or alternative optimal solution

$x_1 = x_2 = x_3 = \frac{5}{2}$, $x_4 = 0$.]

111 Minimize $z = x_1 + x_2$
 subject to $3x_1 + 2x_2 \geq 30$,
 $2x_1 + 3x_2 \geq 30$,
 $x_1 + x_2 \leq 5$,
 $x_1, x_2 \geq 0$.

[Ans. No feasible solution.]

112 Minimize $z = 5x_1 + 11x_2$
 subject to $2x_1 + x_2 \leq 4$,
 $3x_1 + 4x_2 \geq 24$,
 $2x_1 - 3x_2 \geq 6$,
 $x_1, x_2 \geq 0$.

[Ans. No feasible solution.]

[Kalyani Hons., 1981]

113 Minimize $z = x_1 - 2x_2 - 3x_3$
 subject to $-2x_1 + x_2 + 3x_3 = 2$,
 $2x_1 + 3x_2 + 4x_3 = 1$,
 $x_j \geq 0$, $j = 1, 2, 3$.

[Ans. No feasible solution.]

[Bombay, 1974]

114 Show that the following problem has unbounded solution :

Maximize $z = x_1 - x_2 + 2x_3$
 subject to $-x_1 + x_2 + 2x_3 \leq 5$,
 $-2x_1 + 5x_2 - x_3 \geq 10$,
 $2x_1 - x_2 + x_3 \geq 4$,
 $x_1, x_2, x_3 \geq 0$.

[Kalyani M. Sc., 1982]

115 Apply the two-phase process to show that the following L. P. P. has unbounded solution :

Maximize $z = 2x_1 - x_2 + 2x_3$
 subject to $x_1 + x_2 - 3x_3 \leq 8$,
 $4x_1 - x_2 + x_3 \geq 2$,
 $2x_1 + 3x_2 - x_3 \geq 4$,
 $x_1, x_2, x_3 \geq 0$.

[Calcutta Hons., 1982]

116 Use two-phase method to find a B. F. S. of

$x_1 - 2x_2 + x_3 \leq 11$,
 $-4x_1 + x_2 + 2x_3 \geq 3$,
 $2x_1 - x_3 = -1$,
 $x_1, x_2, x_3 \geq 0$.

[Tnpura Hons., 1980]

[Ans. $x_1 = 0$, $x_2 = 1$, $x_3 = 1$.]

- 17 Solve the following equations by simplex method :

$$3x_1 + x_2 = 7,$$

$$x_1 + x_2 = 3.$$

$$[Let \ x_1 = x'_1 - x''_1, \ x_2 = x'_2 - x''_2, \ x'_1, x'_2, x''_1, x''_2 \geq 0.]$$

$$Max \ z = 0.x_1 + 0.x_2 - x_3 - x_4$$

$$subject \ to \ Ax = b,$$

$$where \ A = \begin{bmatrix} 3 & -3 & 1 & -1 & 1 & 0 \\ 1 & -1 & 1 & -1 & 0 & 1 \end{bmatrix}, \ b = \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

$$x = [x_1' \ x_1'' \ x_2' \ x_2'' \ x_3 \ x_4] \geq 0.$$

Start with the basis matrix in which (x_3, x_4) are basic variables and apply phase I simplex method.]

$$[Ans \ x_1 = 2, \ x_2 = 1.]$$

- 18 Solve the following system of equations by simplex method :

$$x_1 - x_3 + 4x_4 = 3,$$

$$2x_1 - x_2 = 3,$$

$$3x_1 - 2x_2 - x_4 = 1,$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

[Delhi, 1975]

$$[Ans \ x_1 = 5, \ x_2 = 7, \ x_3 = 2, \ x_4 = 0.]$$

- 19 Solve the following L. P. P. by simplex method :

$$Maximize \ z = 2x_1 + 5x_2$$

$$subject \ to \ x_1 + 2x_2 \leq 8,$$

$$x_1 \leq 4,$$

$$0 \leq x_2 \leq 3$$

and x_1 is unrestricted in sign.

$$[Ans \ x_1 = 2, \ x_2 = 3; \ z_{max} = 19.]$$

- 20 Solve the following L. P. P. by simplex method :

$$Maximize \ z = 2x_1 + 3x_2$$

$$subject \ to \ -x_1 + 2x_2 \leq 4,$$

$$x_1 + x_2 \leq 6,$$

$$x_1 + 3x_2 \leq 9$$

and x_1, x_2 are unrestricted in sign.

$$[Ans \ x_1 = \frac{3}{2}, \ x_2 = \frac{3}{2}; \ z_{max} = \frac{27}{2}.]$$

- 21 Solve the following L. P. P. by simplex method :

$$Maximize \ z = 3x_1 + 2x_2 + x_3$$

$$subject \ to \ 2x_1 + 5x_2 + x_3 = 12,$$

$$3x_1 + 4x_2 = 11,$$

$$x_1, x_2 \geq 0$$

and x_3 is unrestricted in sign.

$$[Ans \ x_1 = \frac{11}{3}, \ x_2 = 0, \ x_3 = \frac{14}{3}; \ z_{max} = \frac{47}{3}.]$$

- 22 Using simplex method, obtain the inverse of the matrix

$$A = \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix}.$$

[Maximize the objective function

$$z = 0.x_1 + 0.x_2 - x_3 - x_4$$

$$subject \ to \ the \ constraints \ \begin{bmatrix} 3 & 2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix},$$

$$x_1, x_2, x_3, x_4 \geq 0.$$

[Tripura M. Sc., 1980]

(x_1, x_2) are legitimate variables and (x_3, x_4) are the artificial variables, $[6, 4]$ is a dummy column vector.

Then proceed with the phase I simplex method and from the final iteration, the inverse of the matrix A will be given by the column vectors of the initial basis.]

$$[Ans \ A^{-1} = \frac{1}{11} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}]$$

- 23 Using simplex method, obtain the inverse of the matrix

$$A = \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}.$$

$$[Ans \ A^{-1} = \frac{1}{10} \begin{bmatrix} 2 & -4 \\ 1 & 3 \end{bmatrix}]$$

- 24 Using simplex method, obtain the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}.$$

$$[Ans \ A^{-1} = \frac{1}{40} \begin{bmatrix} -4 & 4 & 8 \\ 1 & -11 & 8 \\ 14 & 6 & -8 \end{bmatrix}]$$

- 25 Using simplex method, show that the inverse of the matrix

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ is } \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}.$$